

SHRI G S INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE

SHRI G S INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE													
BTech 1st YEAR SCHEME (Common to all Branches)													
Session July 2025 onwards													
Semester : I										Min 31%		Min 31%	
S.No.	Category	Subject Code	Subject Name	Credits			Hours/Week			Min 31 %		Min 31 %	
				L	T	P	L	T	P	Th	CW	Pr	SW
1	BSC	MA10021	Mathematics for Engineers	2	1	0	2	1	0	70	30	-	-
2	HSMC	HU10181	Understanding Bharat	0	2	1	0	2	2	-	-	60	40
3	ESC	IT10007	Fundamentals of Information Technology and Artificial Intelligence	2	1	2	2	1	4	70	30	60	40
4	ESC	ME10008	Overview of Mechanical Engineering and Graphics	2	0	2	2	0	4	70	30	60	40
5	BSC	PH10009	Applied Physics	2	0	1	2	0	2	70	30	60	40
6	BSC	CH10010	Applied Chemistry	1	0	1	1	0	2	70	30	60	40
			Total	9	4	7	9	4	14	350	150	300	200

SHRI G S INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE													
BTech 1st YEAR SCHEME (Common to all Branches)													
Session July 2025 onwards													
Semester : II									Min 31%		Min 31%		
S.No.	Category	Subject Code	Subject Name	Credits			Hours/Week			Min 31 %		Min 31 %	
				L	T	P	L	T	P	CW	Th	SW	Pr
1	BSC	MA10509	Mathematics for Data Science	2	1	0	2	1	0	70	30	-	-
2	ESC	EE10510	Fundamentals of Electrical and Electronics Engineering	2	1	1	2	1	2	70	30	60	40
3	HSMC	HU10512	Languages for Engineers	1	1	1	1	1	2	70	30	60	40
4	ESC	CE10513	Fundamentals of Civil Engineering and Applied Mechanics	2	1	1	2	1	2	70	30	60	40
5	BSC	PY10514	Biology for Engineers	2	0	1	2	0	2	70	30	60	40
6	ESC	IP10584	Design Thinking and Manufacturing Practices	0	1	2	0	1	4	-	-	60	40
			Total	9	5	6	9	5	12	350	150	300	200
L: Leture (Theory): T: Tutorial: P=Practical (Lab)													

GROUP-I**B. TECH. I YEAR (4YDC) COMMON FOR ALL BRANCHES
MA10021 : MATHEMATICS FOR ENGINEERS**

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
MA10021	Mathematics for Engineers	2	1	0	2	1	0	30	70	-	-

PRE-REQUISITE : NIL**Name of Faculty Coordinator: Dr Rachna Navlakhe****COURSE OBJECTIVES**

The objective of this course is to develop student's analytical and problem-solving skills through foundational concepts of differential and integral calculus, matrices, differential equations, and fuzzy logic, enabling effective application in engineering and scientific domains.

COURSE OUTCOMES

After completing this course student will be able to

CO1:	Identify key contributions of Indian mathematicians from ancient to modern era; acquire the knowledge of partial derivatives and its applications in series expansion and finding maxima and minima.
CO2:	Evaluate Beta and Gamma functions, solve multiple integrals, and apply change of order and variables using Jacobians for mathematical and engineering problems.
CO3:	Analyze and solve matrix-related problems by determining rank, finding inverses, solving linear systems, computing eigenvalues and eigenvectors with applications, and applying the Cayley–Hamilton theorem to obtain

	inverses and powers.
CO4:	Classify differential equations according to features and solve higher order differential equations, with their applications in relevant domain.
CO5:	Differentiate between classical and fuzzy sets, explain and represent membership functions with their properties, perform basic fuzzy set operations, and apply fuzzy set concepts to real-world problems.

COURSE CONTENTS

THEORY

Unit 1	Evolution of Indian Mathematics and Its Contributors: Ancient Era: Baudhayana: Ancient Indian version of the Pythagoras theorem, Aryabhata Varāhamihira, Medieval Era: Brahmagupta Bhaskara II, Madhava of Sangamagrama, Modern era: Ramanujan, Shanti Swaroop Bhatnagar, Harish Chandra. Differential Calculus: Partial derivatives: Definition, Jacobians, Expansion of functions by Taylor's and Maclaurin's series of two variables, Maxima and Minima of functions of two variables.
Unit 2	Integral Calculus: Beta and Gamma functions, Elementary ideas of multiple integrals, Change of order of integration and change of variables in double integrals using Jacobians.
Unit 3	Matrices: Rank of a matrix by Echelon form and Normal form, Inverse of matrices by Gauss- Jordan method, System of linear equations: Solving system of Homogeneous and Non- Homogeneous linear equations, Eigen values, Eigen vectors with their properties and applications, Cayley-Hamilton theorem: Finding inverse and power of a matrix by Cayley Hamilton Theorem.

Unit 4	Ordinary Differential Equations: Ordinary Differential Equations: Linear differential equations with constant coefficients, Second order linear differential equations with variable coefficients, Method of variation of parameters, Application to simple problems.
Unit 5	Fuzzy Sets: Introduction to Fuzzy Sets, Classical sets vs Fuzzy sets, Crisp set characteristics: Membership Functions, Definition and representation, Types of membership functions, Properties of membership functions: Normality Convexity Support, Operations on Fuzzy Sets: Union, Intersection, and Complement, Applications of Fuzzy Sets.

ASSESSMENT

- 1. Internal Assessment for continuous evaluation, mid-term tests, tutorials, class performance, etc. (30%).**
- 2. End semester Theory Exam (70%).**

TEXT BOOKS RECOMMENDED

- 1. B. V. Ramana, Higher Engineering Mathematics, Tata McGraw Hill Publishing Company Ltd., New Delhi, 2006.**
- 2. H. K. Das, Advanced Engineering Mathematics, S. Chand and Company Ltd., New-Delhi, 2009.**

REFERENCE BOOKS

- 1. Paras Ram, Engineering Mathematics through Applications, CBS Publishers & Distributors Pvt. Ltd., Delhi, 2011.**
- 2. M. Ganesh, Introduction to Fuzzy Sets and Fuzzy Logic, PHI Learning Private Limited, 2009.**
- 3. P. Singh, *A Brief History of Mathematics in India*. New Delhi, India: Hindustan Book Agency, 2010.**

CO-PO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1	--	--	--	1	1	1	--	3
CO2	3	3	1	1	--	--	--	1	1	1	--	3
CO3	3	3	2	1	--	--	--	1	1	1	--	3
CO4	3	3	2	1	--	--	--	1	1	1	--	3
CO5	2	3	1	1	--	--	--	1	1	1	--	3

IT10017: FUNDAMENTALS OF INFORMATION TECHNOLOGY & ARTIFICIAL INTELLIGENCE

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
IT10007	Fundamentals of Information Technology & Artificial Intelligence	2	1	2	2	1	4	30	70	40	60

Name of Faculty Coordinator: Prof K K Sharma & Prof Anuradha Rajguru

COURSE OUTCOMES:

After completing the course student should be able to:

CO#1	Explain basics of Information Technology and develop and host the web pages. Translate the algorithms to programs and test and execute the programs and correct syntax and logical errors.
CO#2	Use all programming constructs and write programs using iterative statements, functions, recursion, array, pointers, structures and files.
CO#3	Use all programming constructs and write programs using iterative statements, functions, recursion, array, pointers, structures and files.
CO#4	Describe fundamentals of Artificial Intelligence (AI) and Python Programming.

UNIT 1:	Overview of Information Technology: Information Concepts and Processing, Block Schematic of digital computer and its working, Introduction to computer hardware and software, Overview of Operating System, Basics of computer networks and internet, Information Security basics, IT Act and digital ethics.
UNIT 2:	Flowchart and algorithm, Structure of C program, key words and identifiers, constants and variables, Memory representation of standard structured and enumerated data types, Declarations, scope and life of variables. Various types of operators and expressions, Operator precedence and Associativity. Programming errors and their handling. Data type conversion and type casting, Storage classes, Decision making and Branching: if-else, switch-case. Use of break, continue, default and exit.

UNIT 3:	Use of looping: While-do, for, do-while etc., nesting of loops. Introduction to functions, functions with array, passing parameters to functions, call by value, call by reference, recursive functions. Introduction to Arrays: Structure, and union, nesting of structures, structure passing in function, Array of structures; Pointers: declaration, applications, Introduction to dynamic memory allocation, Use of pointers in self-referential structures, Files: File I/O functions
UNIT 4:	Introduction to Databases: Definition, purpose of databases, Real-life examples and applications, Data vs. Information, File System vs. Database System, Limitations of traditional file systems, Advantages of DBMS (data integrity, security, redundancy reduction), Relational Databases, Definition and functions, Relational Model Basics, Tables, rows (tuples), columns (attributes), Keys: Primary, Foreign key, Basic SQL syntax, DDL: CREATE, DROP, ALTER, DML: SELECT, INSERT, UPDATE, DELETE
UNIT 5:	Introduction to Artificial Intelligence: Definition, brief history, and scope, Problem Solving and Search with AI, Evolution of AI Technologies, Introduction to Neural Network, Machine Learning, Deep Learning, Generative AI; AI tools and platforms, Legal, ethical, and social implications of AI.

Text Books

1. E. Balagurusamy, "Programming in ANSI C", Sevenths Edition, Tata McGraw Hill, 2017.
2. Reema Thareja, "Programming in C", Second Edition , Oxford publication, 2016.
3. W. Kernighan and Dennis M. Ritchie, "The C Programming Language", Pearson , 2015.
4. Russel, S., and Norvig, P., "Artificial Intelligence: A Modern Approach", 4th Edition, 2020, Pearson.
5. Elaine Rich, Kevin Knight, Shivashankar B. Nair, "Artificial Intelligence", 3rd Edition, 2008, McGraw-Hill International.
6. Arockia Mary P, Problem Solving and Python Programming, Shanlax Publications, 2021.

Reference books

1. Matthias Felleisen, Robert Bruce Findler, Mathew Flatt, Shriram Krishnamurthi, "How to Design Programs: An Introduction to Programming and Computing", Second Edition, MIT Press, 2018.
2. M. Chandwani, A. Jain and N.S. Chaudhary, "Elements of Computer Science", Jain Publishers 1997.
3. B.S. Gottfried, "Programming with C", 3rd edition, Tata McGraw Hill, 2018
4. Problem Solving and Program Design in C, by Jeri R. Hanly, Elliot B. Koffman, Pearson Addison-Wesley, 2006.
5. W. Patterson, 'Introduction to Artificial Intelligence and Expert Systems', 5th Edition, 2003, Prentice Hall of India.
6. Z. Shaw, LEARN PYTHON 3 THE HARD WAY, Addison-Wesley, 2017.
7. Lemay. Laura, Rafe Colburn, Jennifer Kyrnin, Mastering HTML, CSS & JavaScript Web, BPB Publication, 2016

ME10008: OVERVIEW OF MECHANICAL ENGINEERING AND GRAPHICS

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
ME10008	Overview of Mechanical Engineering and Graphics	2	0	2	2	0	4	30	70	40	60

Name of Faculty Coordinator:

Course Outcomes:

CO's	Statement
CO1	Explain the types of materials and their properties for engineering applications
CO2	Analyze the fundamentals of links and pair for different mechanism of machine
CO3	Enhancement of fundamental knowledge of a variety of practical fluid flow.
CO4	Analyze the efficiency of I.C. Engine and working of automobile components
CO5	Distinguish standard drawing conventions, apply the concept of orthographic projection of line, plane, solid and section of solid.

Articulation Matrix

COs/POs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2
CO1	3	2	-	2	-	-	-	-	-	-	-	1	3	2
CO2	3	2	-	-	-	-	-	-	-	-	-	1	3	2
CO3	3	3	-	2	-	-	-	-	-	-	-	1	3	2
CO4	3	2	2	-	-	-	-	-	-	-	-	1	2	2
CO5	2	-	2	-	3	-	-	-	-	2	-	1	3	3

COURSE CONTENTS

UNIT 1:	Engineering Materials and Properties: Introduction, properties, normal tensile shear and thermal stress, stress-strain diagrams for ductile and brittle materials, elastic constants and relationship between elastic constant.
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UNIT 2:	Mechanism and Machines: Types of motion, links, kinematic pair, types of joints, degree of freedom, classification of kinematic pairs, kinematic chain, linkage, mechanism and structure, inversions of basic mechanisms (four- bar chain, single slider and double slider crank mechanisms) and their applications. Hook's joint, Types of gears and gear trains.
UNIT 3:	Fluid Kinematics: Lagrangian & Eulerian Method, Types of Flow and Types of motion, local and convective acceleration, continuity equation.
UNIT 4:	Introduction of IC Engines and Automobile: Classification of IC engines, terminology used in I C engine; Otto & Diesel Cycle; their air standard efficiencies. Principal parts of IC engine, their functions and working. Introduction of electrical vehicle, components of automobile and their function like brakes, clutch, gear etc.
UNIT 5:	Engineering Graphics: Principles of Engineering Graphics and their significance, usage of Drawing instruments, lines, lettering and dimensioning, Introduction of Projection of Line, Plane, Solid and Section of Solid, overview of orthographic and isometric projection.

ASSESSMENT:

The following methods shall be adopted for the assessment of this course;

- 1. Theory Examination (70 marks) on the basis of end term theory paper examination (from Units 1 to 5)**
- 2. Class Work (30 marks) on the basis of regular evaluation of assignments, two mid semester tests and class attendance.**
- 3. Sessional Work (40 marks) on the basis of assignments, internal viva and attendance. Student are require to submit at least five imperial drawing sheets/ computer aided drawing print.**
- 4. Practical examination (60 marks) on the basis of evaluating practical knowledge, Quiz and viva voce.**

TEXT BOOKS RECOMMENDED:

- 1. Kumar D.S., Fluid Mechanics, S K Kataria Publication, 2003**
- 2. Ganeshan, I. C. Engine, Tata McGraw Publication, 2012**
- 3. Bansal R K, Strength of Materials, Laxmi Publication**
- 4. Rattan S. S. Theory of Machine, Tata McGraw Hill.**

List of Experiments:-

Sheet 1:- Orthographic Projection

Sheet 2:- Scales and Projection of Line

Sheet 3:- Projection of Planes

Sheet 4:- Projection and section of Solids

Sheet 5:- Isometric Projection

TEXT BOOKS RECOMMENDED:

- 1. N.D. Bhatt and V.M. Panchal, Engineering Drawing Plane and Solid Geometry, Charotar Publishing House**
- 2. Agrawal Basant and Agrawal C.M. (2018), Engineering Drawing, McGraw Hill Publishing.**

PH10009: APPLIED PHYSICS

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
PH10009	Applied Physics for Engineers	2	0	1	2	0	2	-	-	-	-

Name of Faculty Coordinator:

COURSE OBJECTIVES

CO#1	To provide knowledge and understanding of basic, applied, and modern physics. / मूल, व्यावहारिक और आधुनिक भौतिकी का ज्ञान और समझ क्षमता प्रदान करना ।
CO#2	To generate attitude and interest to solve problems at macro, and micro level systems. / स्थूल, सूक्ष्म व नैनो स्तर की प्रणालियों में प्रश्नो को हल करने के लिए दृष्टिकोण और रुचि उत्पन्न करना ।
CO#3	To understand the classification and properties of both basic and applied physics at nanoscale systems / मूल और अनुप्रयुक्त भौतिकी दोनों के वर्गीकरण और गुणों नैनो स्तर पर समझना ।
CO#4	To update the knowledge of physics tools, instruments, and techniques incorporating human values and safety measures. / मानव मूल्यों और सुरक्षा उपायों को शामिल करते हुए भौतिकी के उपकरणों और तकनीकों के ज्ञान को सामयिक बनाना ।
CO#5	To identify, conduct, formulate and solve engineering problems with the applied knowledge of Physics. / भौतिकी के मूल और व्यावहारिक ज्ञान के साथ अभियांत्रिकी प्रश्नो की पहचान, संचालन, सूत्रीकरण और समाधान करना ।

COURSE CONTENTS / अध्ययन की विषयवस्तु

UNIT 1	Lasers: Photon absorption & emission, role of S N Bose, components of lasers, optical resonator, Einstein's A & B coefficients, Population inversion, Ruby and He-Ne lasers, contribution of CKN Patel and M L Bhaumik, लेजर: तात्क्षणिक और उद्दीपित उत्सर्जन, एस एन बोस की भूमिका, , लेजर के घटक, प्रकाशीय अनुनादक, आइंस्टीन के A और B गुणांक, समष्टि प्रतिलोमन, He-Ne लेजर, सीकेएन पटेल और एमएल भौमिक का योगदान, ।
UNIT 2	Fiber Optics: NS Kapany's fiber optics, Classification of fibers, acceptance angle, numerical aperture, V-number, losses, FO sensors, FO communication Systems. तंतु (फाइबर) प्रकाशिकी: एनएस कपानी के फाइबर ऑप्टिक्स, तंतु (फाइबर) के वर्गीकरण, स्वीकृति कोण, संख्यात्मक छिद्र, V-नंबर, क्षीणन, प्रकाशीय तंतु सुग्राही, प्रकाशीय तंतु संचार प्रणाली।
UNIT 3	Special Theory of relativity (STR): Postulate of STR, time dilation, length contraction, twin paradox, Relativistic velocity of particles, Relativistic variation of mass, mass-energy relation. विशिष्ट आपेक्षिकता सिद्धांत: आपेक्षिकता सिद्धांत, समय फैलाव, लंबाई संकुचन, जुड़वां विरोधाभास, कणों का आपेक्षिक वेग, द्रव्यमान का आपेक्षिक परिवर्तन, द्रव्यमान-ऊर्जा संबंध ।
UNIT 4	Quantum Theory: Theory of superposition, Planck's radiation formula, S Chandrashekar and idea of black hole the Bose in Boson, Compton's effect, pair production, de Broglie's and Heisenberg's hypotheses, Schrodinger's wave equation, interpretation of wave function, particle in a 1-D potential well. क्वांटम सिद्धांत: अध्यारोपण का सिद्धांत, प्लैंक का विकिरण सूत्र, एस चंद्रशेखर और ब्लैक होल का विचार बोसॉन में बोस, कॉम्पटन का प्रभाव, डी ब्रोगली की पदार्थ तरंगों की अवधारणा, हाइजेनबर्ग का अनिश्चितता सिद्धांत, श्रोडिंगर का तरंग समीकरण, तरंग फलन की भौतिक व्याख्या, एक-आयामी विभव कूप में कण की परिकल्पना।
UNIT 5	Quantum Computation: Classical Gates and operations, AND, OR, NAND, XOR Gates and operations. Unitary operations, Pauli Matrices, Quantum Logics: Hadamard, Pauli-X, Y, Z and R, CNOT, Swap Gates and their unitary operations, influence of ECG Sudharshan and GS Agarwal. क्वांटम कम्प्यूटेशन: शास्त्रीय द्वार और संचालन, और, या, नंद, एक्सओआर गेट्स। क्वांटम गेट्स और संचालन। एकात्मक

	संचालन, पाउली मैट्रिसेस, क्वांटम लॉजिक्स: हैडमार्ड, पाउली-एक्स, वाई, जेड और आर, सीएनओटी, स्वैप गेट्स और उनके एकात्मक संचालन, ईसीजी सुदर्शन और जीएस अग्रवाल का प्रभाव ।
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Text Books

1. A. Beiser, S. Mahajan and S. R. Choudhary: Concepts of Modern Physics, 6th Ed., (Tata McGraw Hill, New Delhi) 2012.
2. A. Ghatak: Optics, 4th Edition, (Tata McGraw-Hill, New Delhi) 2009.
3. Fundamental Quantum Mechanics for Engineers, Leon van Dommelen, 15 Jun 2012 Version 5.55 alpha, (Ebook: <http://www.eng.fsu.edu/~dommelen/quantum/>).

Reference books

4. N. Subramanyam and B. Lal: A Text book of Optics, (S. Chand, New Delhi) 2010.
5. H. K. Malik and A. K. Singh : Engineering Physics (Tata McGraw Hill New Delhi) 2010.
6. W. H. Hayt: Engineering Electromagnetic, 5th Ed. (Tata-McGraw Hill, New Delhi) 1995.
7. V Sahani: Quantum Computing (Tata McGraw-Hill, New Delhi) 2009

PH10009: APPLIED PHYSICS LABORATORY/ अनुप्रयुक्त भौतिकी प्रयोगशाला

LIST OF EXPERIMENTS / प्रयोगों की सूची

1. Error Analysis in Physics Laboratory. / भौतिकी प्रयोगशाला में त्रुटि विश्लेषण ।
2. To determine the wavelength of sodium light by Newton's rings method. / न्यूटन वलय विधि द्वारा सोडियम प्रकाश की तरंगदैर्घ्य ज्ञात करना ।
3. To study the variation of magnetic field along the axis of a circular coil carrying current and to calculate the radius of the coil. / एक वृत्ताकार कुंडली के अक्ष के अनुदिश चुंबकीय क्षेत्र की भिन्नता का अध्ययन करना तथा कुंडली की त्रिज्या की गणना करना ।

4. To measure the numerical aperture of given optical fiber. / दिए गए ऑप्टिकल फाइबर के संख्यात्मक छिद्र की गणना करना ।
5. To measure Planck's constant using light emitting diodes (LED) of various colors and to understand work function. / विभिन्न रंगों के प्रकाश उत्सर्जक डायोड (LED) का उपयोग करके प्लैंक के नियतांक को मापना और कार्यफलन को समझना ।
6. To study working of laser using PhET Module./PhET मॉड्यूल का उपयोग करके लेजर की कार्यप्रणाली का अध्ययन करना ।
7. To study the law of dispersion by a triangular prism using raytrace. / रे-ट्रेस का उपयोग करके, एक त्रिभुजाकार प्रिज्म द्वारा प्रकीर्णन का अध्ययन करना ।
8. To understand and confirm Heisenberg's uncertainty principle using single slit diffraction. / एकल रेखाछिद्र विवर्तन का उपयोग करते हुए हाइजेनबर्ग के अनिश्चितता सिद्धांत को समझना और पुष्टि करना ।
9. To determine the wavelength of prominent spectral lines of mercury light by a plane transmission grating using normal incidence. / अभिलंब आपतन का उपयोग करते हुए एक समतल पारगमन ग्रेटिंग द्वारा प्रकाश की प्रमुख वर्णक्रमीय रेखाओं की तरंग दैर्घ्य निर्धारित करना ।
10. To measure the charge to mass ratio of an electron using Thomson method and to find the polarity of the charge of electron. / थॉमसन विधि का उपयोग करके एक इलेक्ट्रॉन के आवेश और द्रव्यमान के अनुपात को मापना तथा इलेक्ट्रॉन के आवेश की ध्रुवता ज्ञात करना।
11. To verify Kirchhoff's law and Inverse-square law in thermodynamics using Leslie's cube. / लेस्ली के क्यूब का उपयोग करके, ऊष्मागतिकी के किरचॉफ के नियम और व्युत्क्रम-वर्ग नियम को सत्यापित करना।

REFERENCE MANUAL / संदर्भ तालिका

Download Lab Manual from the website (<https://sgsits.ac.in>) of Shri G. S. Institute of Technology and Science, Indore. Consult the references given in each chapter of the manual for more information. Download Tutorial Sheets from

<http://dx.doi.org/10.13140/RG.2.2.28342.63044>. / श्री जी. एस. प्रौद्योगिकी एवं विज्ञान संस्थान, इंदौर की वेबसाइट (<https://sgsits.ac.in>) से प्रयोगशाला पुस्तिका डाउनलोड करें। अधिक जानकारी के लिए पुस्तिका के प्रत्येक अध्याय में दिए गए संदर्भों को देखें। <http://dx.doi.org/10.13140/RG.2.2.28342.63044> से अनुशिक्षण पत्रक डाउनलोड करें।

Course Outcome

At the end of one-semester course, the students are armed with the / एक अर्ध वार्षिक सत्र के पाठ्यक्रम के अध्ययन से, छात्र सक्षम होंगे

CO1 knowledge of multiphysics to understand and solve basic engineering problems. / बहु-भौतिकी का ज्ञान तथा मूल अभियांत्रिकी प्रश्नों को समझने और हल करने में

CO2 logical attitude towards the solution of engineering problems with modern physics implementation. / आधुनिक भौतिकी कार्यान्वयन के साथ अभियांत्रिकी प्रश्नों के समाधान के लिए तार्किक दृष्टिकोण में

CO3 ability to use modern techniques and tools including software involving advanced physics to engineering subjects. / अभियांत्रिकी विषयों के लिए उन्नत भौतिकी से जुड़े सॉफ्टवेयर सहित आधुनिक तकनीकों और उपकरणों का उपयोग करने की क्षमता में।

CO4 comprehensive applications of engineering physics of laser & fiber optics theory of relativity, quantum mechanics, / लेजर और फाइबर ऑप्टिक्स, सापेक्षता के सिद्धांत, क्वांटम यांत्रिकी के इंजीनियरिंग भौतिकी के व्यापक अनुप्रयोग।

CO5 creative engineering solutions for society incorporating human values and safety measures with the applied physics knowledge. / अनुप्रयुक्त भौतिकी ज्ञान के साथ मानवीय मूल्यों और सुरक्षा उपायों को शामिल करते हुए समाज के लिए रचनात्मक अभियांत्रिकी समाधान करने में।

CH10010: CHEMISTRY FOR ENGINEERS

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
CH10010	Applied Chemistry for Engineers	1	0	1	1	0	2	30	70	40	60

Name of Faculty Coordinator: Prof Nitish Gupta

Program Outcomes (POs):

POs describe what students should know and be able to do at the end of the programme. POs are to be in line with the graduate attributes as specified in the Washington Accord. POs are to be specific, measurable and achievable. **NBA has defined 12 POs.**

Graduates will be able to achieve:

1.	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, & an engineering specialization to the solution of complex engineering problems.
2.	Problem analysis: Identify, formulate, review research literature, & analyze complex engineering problems reaching substantiated conclusions using 1st principles of math., natural sciences, & eng. sciences.
3.	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4.	Conduct investigations of complex problems: Use research-based knowledge & research methods including design of experiments, analysis & interpretation of data, & synthesis of the information to provide valid conclusions.
5.	Modern tool usage: Create, select, & apply appropriate techniques, resources, & modern engineering & IT tools including prediction & modeling to complex engineering activities with an understanding of the limitations.
6.	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7.	Environment & sustainability: Understand the impact of the professional engineering solutions in societal & environmental contexts, & demonstrate the knowledge of, & need for sustainable development.
8.	Ethics: Apply ethical principles & commit to professional ethics & responsibilities and norms of the engineering practice.
9.	Individual & team work: Function effectively as an individual, & as a member or leader in diverse teams, and in multidisciplinary settings.
10.	Communication: Communicate effectively on complex engineering activities with the engineering community & with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give & receive clear instructions.
11.	Project management and finance: Demonstrate knowledge & understanding of the engineering & management principles & apply these to one's own work, as a member & leader in a team, to manage projects & in multidisciplinary environments.
12.	Life-long learning: Recognize the need for, & have the preparation and ability to engage

13. in independent and life-long learning in the broadest context of technological change.

COURSE OBJECTIVES

1.	To deliver a comprehensive understanding of fundamental, applied, and advanced concepts in chemistry./ रसायन विज्ञान की मौलिक, अनुप्रयुक्त और उन्नत अवधारणाओं की समग्र समझ प्रदान करना।
2.	To recognize, design, analyze, and resolve engineering challenges using applied knowledge of chemistry./ रसायन विज्ञान के अनुप्रयुक्त ज्ञान का उपयोग करते हुए इंजीनियरिंग से जुड़ी समस्याओं की पहचान करना, उनका डिज़ाइन तैयार करना, विश्लेषण करना और समाधान करना।
3.	To Build student's proficiency to effectively use analytical techniques for material characterization across multiple engineering applications./ विभिन्न इंजीनियरिंग अनुप्रयोगों में सामग्री के चरित्रण हेतु विश्लेषणात्मक तकनीकों का प्रभावी ढंग से उपयोग करने की छात्रों की दक्षता का विकास करना।

4. To upgrade knowledge of chemistry-related techniques, ensuring adherence to human values and safety standards./ रसायन विज्ञान से संबंधित तकनीकों के ज्ञान को अद्यतन करना, साथ ही मानव मूल्यों और सुरक्षा मानकों का पालन सुनिश्चित करना।

Course Content:

UNIT-I	Water: Analytical aspects, treatment and industrial applications/ जल: विश्लेषणात्मक पक्ष, उपचार और औद्योगिक अनुप्रयोग Introduction, Source, Types of impurities and their effects, Hardness, its expression and determination, BIS specifications for Industrial and domestic water, overview of Boiler troubles and softening methods, Numericals./ परिचय, स्रोत, अशुद्धियों के प्रकार और उनके प्रभाव, कठोरता, उसकी अभिव्यक्ति और निर्धारण, औद्योगिक और घरेलू जल के लिए बीआईएस विनिर्देश, बॉयलर से संबंधित समस्याओं का अवलोकन और जल मृदुकरण विधियाँ, संख्यात्मक प्रश्न।
Unit-II	Introduction to lubricants , fuels & Combustion/ स्नेहक, ईंधन एवं दहन की भूमिका (a) Lubricants: Introduction, Types of lubricants and principles of lubrication's, properties (test) of lubricants, numericals./ स्नेहक (Lubricants): परिचय, स्नेहकों के प्रकार और स्नेहन के सिद्धांत, स्नेहकों के गुणधर्म (परीक्षण सहित), संख्यात्मक प्रश्न। (b) Fuel and combustion, numericals / ईंधन एवं दहन (Fuel and Combustion): ईंधन और दहन की मूल बातें, संख्यात्मक प्रश्न। Assessment (Theory)/ मूल्यांकन: Class test, assignments/quiz, attendance and end semester theory exam/ मध्य सत्र परीक्षा, असाइनमेंट/प्रश्नोत्तरी, उपस्थिति और सेमेस्टर अंत परीक्षा।

Books & References Recommended/ पुस्तकें एवं अनुशंसित संदर्भ पुस्तकें::

Text Books/ पाठ्य पुस्तकें:

1. Dara S.S., Engineering Chemistry (S. Chand publishing)/ **एस.एस.दारा**, इंजीनियरिंग केमिस्ट्री (चंद पब्लिशिंग एस)
2. P.C. Jain and M Jain, Engineering Chemistry, 15th Edition, Dhanpat Rai publishing Co./ **पी.सी. जैन एवं एम. जैन**, इंजीनियरिंग केमिस्ट्री, 15वां संस्करण, धनपत राय पब्लिशिंग कंपनी
3. JC Kuriacose and J Rajaram, Chemistry in Engineering and Technology, Vol-I & II, Tata Mcgraw Hill Education Pvt. Limited. **जे.सी. कुरियाकोस एवं जे. राजाराम**, केमिस्ट्री इन इंजीनियरिंग एंड टेक्नोलॉजी, खंड-I एवं II, टाटा मैकग्रा हिल एजुकेशन प्रा. लि.

Reference Books/ संदर्भ पुस्तकें:

1. Palanna O.G, Engineering Chemistry (Mc Graw Hill)/ **ओ. जी. पलन्ना**, इंजीनियरिंग केमिस्ट्री (मैकग्रा हिल)
2. Ambasta B.K., Chemistry for Engineers (University Science Press/ **बी. के. अंबस्ता**, इंजीनियरों के लिए रसायन विज्ञान (यूनिवर्सिटी साइंस प्रेस)

B.Tech. I Year Chemistry Semester-A&B CH10010: APPLIED CHEMISTRY Laboratory

Course Outcomes (COs): For Laboratory Course:

This laboratory course will illustrate the principles of chemistry relevant to the study of science and engineering./ यह प्रयोगशाला पाठ्यक्रम विज्ञान और अभियांत्रिकी अध्ययन से संबंधित रसायन शास्त्र के सिद्धांतों को स्पष्ट करेगा।

The students will learn to:

CO1: Apply specifications, testing, and treatment of water for industrial and domestic use./ औद्योगिक और घरेलू उपयोग के लिए पानी की विशेषताओं, परीक्षण और उपचार को लागू करना।

CO2: Explain and compute different types of classical and alternative fuels./ विभिन्न प्रकार के पारंपरिक और वैकल्पिक ईंधनों की व्याख्या और गणना करना।

CO3: Illustrate concepts, manufacturing, and applications of different types of industrially important materials and their maintenance./ विभिन्न प्रकार की औद्योगिक रूप से महत्वपूर्ण सामग्रियों की अवधारणाओं, विनिर्माण, अनुप्रयोगों और उनके रखरखाव को चित्रित करना।

CO4: Explain the applications of spectral and analytical techniques in qualitative and quantitative analyses./गुणात्मक और मात्रात्मक विश्लेषण में वर्णक्रमीय और विश्लेषणात्मक तकनीकों के अनुप्रयोगों को समझना।

CO5: Relate the ethics behind applying chemical methods in industries and day-to-day use./उद्योगों में रासायनिक विधियों को लागू करने और दैनिक उपयोग में उनकी नैतिकता को समझना।

List of Experiments for B.Tech. I year, Applied Chemistry/ बी.टेक. प्रथम वर्ष, रसायन विज्ञान हेतु प्रयोगों की सूची

1. Determination of the viscosity of lubricating oil by Redwood Viscometer No.1/ Redwood Viscometer No.2 (at five different temperatures)./ रेडवुड विस्कोमीटर क्रमांक 1 / रेडवुड विस्कोमीटर क्रमांक 2 द्वारा स्नेहक तेल की श्यानता (viscosity) का पाँच विभिन्न तापमानों पर निर्धारण।
2. Determination of flash point of given oil by Abel's apparatus/ Pensky Martin's apparatus./ एबेल्स उपकरण / पेंसकी मार्टिन उपकरण द्वारा दिए गए तेल के फ्लैश पॉइंट का निर्धारण।
3. Determination of percentage of moisture in a coal sample/ Total Solids in a water sample (Gravimetric analysis)./ कोयले के नमूने में आर्द्रता प्रतिशत / जल नमूने में कुल ठोस पदार्थ का निर्धारण (गुरुत्व विश्लेषण विधि से)।
4. Determination of Steam Emulsification number (SEN) of a given lubricating oil sample./ दिए गए स्नेहक तेल नमूने का स्टीम इमल्सिफिकेशन संख्या (SEN) का निर्धारण।
5. Determination of Aniline Point of a given oil sample./ दिए गए तेल नमूने का एनीलीन बिंदु (Aniline Point) का निर्धारण।
6. Determination of Drop Point/Penetration no. of a given semi-solid lubricant./ दिए गए अर्ध-ठोस स्नेहक का ड्रॉप प्वाइंट / पेनेट्रेशन नंबर का निर्धारण।
7. Determination of acid value of an oil sample./ तेल नमूने का अम्ल मान (Acid Value) का निर्धारण।
8. Determination of hardness of water sample by EDTA method./ EDTA विधि द्वारा जल नमूने की कठोरता का निर्धारण।
9. Determination of percentage purity of iron alloy by internal indicator method./ आंतरिक संकेतक विधि से लोहे की मिश्र धातु की प्रतिशत शुद्धता का निर्धारण।

10. Determination of carbonates, bicarbonates and total alkalinity of a water sample./ जल नमूने में कार्बोनेट्स, बाइकार्बोनेट्स एवं कुल क्षारीयता का निर्धारण।

Text Book / प्रायोगिक रसायन विज्ञान हेतु पुस्तकें:

1. A Textbook on Experiments and Calculations in Engineering Chemistry, SS Dara, S. Chand Publisher/ इंजीनियरिंग रसायन विज्ञान में प्रयोग और गणनाएँ – एस. एस. दारा, एस. चंद प्रकाशक।
2. Practical Engineering chemistry, Archana Thakur, Narosa Publishing house, New Delhi/ प्रायोगिक इंजीनियरिंग रसायन – अर्चना ठाकुर, नोरोसा पब्लिशिंग हाउस, नई दिल्ली।

Reference book

1. Vogel's Text Book of Quantitative Analysis, Ed. GH Jeffery, J Bassett, J. Mendham and RC Denny, Longmans, London./ वोगेल का क्वांटिटेटिव एनालिसिस पर पाठ्यपुस्तक, संपादक: जी.एच. जेफरी, जे. बैसेट, जे. मेंडहम और आर.सी. डेनी, लॉन्गमैन्स, लंदन।
2. Download the chemistry theory and practical related study materials: [Shri G. S. Institute of Technology and Science, Indore - Downloads](#)

Course Outcomes (COs): For Theory Course

Course outcomes are the statements of what a student should know, understand and/or be able to demonstrate after completion of a course.

The course will enable student to:

- CO1** Apply Specifications, testing and treatment of water for industrial and domestic use./ औद्योगिक और घरेलू उपयोग के लिए जल के विनिर्देशों, परीक्षण और उपचार को लागू करना।
- CO2** Implement appropriate lubrication strategies in the design and maintenance of engineering systems/ इंजीनियरिंग प्रणालियों के डिज़ाइन और अनुरक्षण में उपयुक्त स्नेहन रणनीतियों को लागू करना।
- CO3** Describe the characteristics of fuels and the fundamentals of combustion processes./ ईंधनों की विशेषताओं और दहन प्रक्रियाओं के मूल सिद्धांतों का वर्णन करना।

10010)/ बी.टेक प्रथम वर्ष रसायन शास्त्र (थ्योरी) हेतु CO-PO मैपिंग शीट

[illegible]

CO PO Mapping Sheet for B. Tech. First Year APPLIED Chemistry Practical (CHXXXX)/ बी.टेक प्रथम वर्ष रसायन शास्त्र (प्रयोगात्मक) हेतु CO-PO मैपिंग शीट

[illegible]

GROUP-II**B. TECH. I YEAR (4YDC) COMMON FOR ALL BRANCHES****MA10509: MATHEMATICS FOR DATA SCIENCE**

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
MA10509	Mathematics for Data Science	2	1	0	2	1	0	-	-	-	-

COURSE OBJECTIVES

This course focuses on the applicability of preliminary mathematical, statistical, and modelling techniques aims to provide students with the know-how and usage of data science to be aligned with the current industry needs.

COURSE OUTCOMES

After completing this course student will be able to

CO1	Develop the capability to perceive and apply concepts of statistics and probability principles relevant to data science.
CO2	Demonstrate the usage of random variables and its applications for analyzing and visualizing the data patterns and distributions.
CO3	Cater to the ability to apply and validate different descriptive and inferential statistical and modeling techniques for data analysis using real datasets conforming to appropriate model fits
CO4	Determine and assess the applicability of key concepts in predictive modelling techniques linked to correlation and regression analyses.
CO5	Apply appropriate data science tools and techniques to analyze real-world datasets and case studies.

COURSE CONTENTS

THEORY

Unit 1	Introductory Statistics and Probability Theory: Introduction of Data – Conceptualization, Scale Classification (Nominal, Categorical, Ordinal, Interval, and Ratio), Population and Sample, Measures of Central Tendency – Mean, Mode and Median, Measures of Dispersion - Range, Quartile, Mean Absolute Deviation, Mean Squared Deviation and Standard Deviation. Random Experiments, Sample Space, and Algebra of Events, Axioms of the Probability Theory Bayes' Theorem.
Unit 2	Random Variables: Discrete and Continuous random variables, Probability Mass Function, Probability Density Function, Cumulative Distribution Function, Mathematical Expectation: mean and variance of random variables. Probabilistic Distributions – Binomial, Poisson, and Normal Distributions (Properties and Standard Normal Variate).
Unit 3	Descriptive and Inferential Statistics: Data Visualization Tools/Techniques – tables, bar charts, histograms, pie charts, box plots. Inferential Statistics: Sampling methods and sampling distributions, Central Limit Theorem, Confidence intervals for means and proportions, Hypothesis testing basics (null/alternative hypotheses, p-value, and significance level).
Unit 4	Correlation and Regression: Correlation Analysis – Pearson Correlation Coefficient, Spearman Rank Correlation, Regression, Types of Regression Models, Simple Linear Regression–Model Building, and Estimation of Parameters Using Ordinary Least Squares, Interpretation and Validation of Simple Linear Regression Coefficients, Coefficient of Determination (R^2).
Unit 5	Data Science Tools and Case Studies: Introduction to data tools, MS-Excel for data analysis, R-Programming for statistical computing, MATLAB tools for engineering applications; and Case studies.

ASSESSMENT

- 1. Internal Assessment for continuous evaluation, mid-term tests, tutorials, class performance, etc. (30%).**
- 2. End semester Theory Exam (70%).**

TEXT BOOKS

- 1. VeeraRajan T, Statistics, Probability and Random Process, 2nd Edition, Tata McGraw Hill Publishing Company Ltd., New Delhi 2003**
- 2. H. K. Das, Advanced Engineering Mathematics, S. Chand and Company Ltd., New-Delhi, 2009.**

REFERENCE

1. Saltz, J.S., and Stanton, J.M., (2018), An Introduction to Data Science, Sage Publications (California, US)
2. M. Ray, H. S. Sharma and S Chaudhary, Mathematical Statistics, Ram Prasad and Sons, Agra, 2004.

CO-PO MAPPING

CO	PO1	PO2	PO3	PO4	PO 5	PO 6	PO 7	PO8	PO9	PO1 0	PO1 1	PO12
CO1	3	3	3	1	1	--	--	1	1	1	--	3
CO2	3	3	3	1	1	--	--	1	1	1	--	3
CO3	3	3	3	2	1	--	--	1	1	1	--	3
CO4	3	3	3	2	1	--	--	1	1	1	--	3
CO5	3	3	2	1	3	--	--	1	3	2	--	3

Teacher Assigned to the Course : Dr. Smita Verma

CE10513: FUNDAMENTALS OF CIVIL ENGINEERING AND APPLIED MECHANICS

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
CE10513	Fundamentals Civil Engineering and Applied Mechanics	2	1	1	2	1	2	30	70	40	60

Name of Faculty Coordinator:

PRE-REQUISITE: XII standard (HSSC)

COURSE OBJECTIVES:

1	To study the concepts of various force systems, equilibrium of force system, concept of trusses
2	To learn the concepts of centroids, centers of gravity and moments of inertia of simple and complex geometrical shape.
3	To learn the behavior of beams, frames and trusses under different loading conditions and to determine the shear force and bending moment diagram for beams subjected to simple point loads.
4	To learn the scope, historical development, various sub-branches of civil engineering, basic materials, and modern technological applications in civil engineering. Introduction to building materials
5	To develop basic understanding of surveying principles and the use of modern instruments in geomatics

COURSE OUTCOMES:

The students shall be able to

1	Identify and analyze system of forces, calculate support reactions for beams subjected to various types of loadings and determine forces in members of trusses.
2	Determine the Center of Gravity and moment of Inertia of a given Plane Lamina & Solid Bodies.
3	Determine and analyze the shear force, bending moment of beams and analyze the trusses and problems related to frictions.
4	Explain the historical background, scope, and various branches of civil

	engineering, including the role of materials and ICT in the field.
5	Apply the basic principles of plane surveying using conventional and modern instruments for measurement and field practices.

COURSE CONTENTS: THEORY

UNIT-1	Forces and Equilibrium: Graphical and Analytical Treatment of Concurrent and non-concurrent Co-planer forces, free body Diagram, Force Diagram and Bow's notations, Application of Equilibrium Concepts: Analysis of plane Trusses: Method of joints, Method of Sections.
UNIT-2	Centre of Gravity and Moment of Inertia: Centroid and Centre of Gravity, Moment of Inertia of Area and Mass, Radius of Gyration, Introduction to Product of Inertia and Principle Axes.
UNIT-3	Support Reaction and analysis of Beams: Types of Beams: Simply Supported Beam, Overhanging Beam, and Cantilever Beam. Types of Supports of a Beam or Frame: Roller, Hinged and Fixed Supports. Load on the Beam and Frame: Different Types of Loading. Support Reaction of a Beam or Frame. Introductions to shear force and Bending Moment for Cantilever & simply supported beam with concentrated, distributed load and Couple.
UNIT-4	Introduction to Civil Engineering: Historical perspective of Civil Engineering, Role of Civil Engineering in infrastructure building of nation and the world. Introduction of various branches of Civil Engineering, Significance and major aspects of each branch and their correlation,
UNIT-5	Introduction to Geomatics / Plane Surveying, Introduction to the principles of plane surveying, Linear and angular measurements in plane surveying. Vertical distance measurement and leveling, Contours and its properties, Examples of field practices. Modern surveying equipment and their applications. Applications of Electronics, Software and ICT in Civil Engineering. Introduction of basic civil engineering materials and their applications.

ASSESSMENT:

Continuous:

1. Class work will be based on following
 - Submission of assignments based on the syllabus of Engineering Mechanics and Surveying.
 - Two midterm tests in a semester and a makeup test if required, Evaluation of Practical calculations & drawing sheets, internal submission and Viva Voice examination by internal examiner during the semester end.
2. Semester-end: Theory examination of 3 Hours duration.

Laboratory Work

OBJECTIVES:

1. To verify the law of polygon of forces using a universal force table by demonstrating equilibrium of concurrent coplanar forces.
2. To determine the support reactions in a simply supported beam using the graphical method and apply Lami's theorem to find unknown forces in a system of three concurrent forces
3. To acquire practical knowledge on handling basic survey equipments..
4. To possess knowledge about chain, compass surveying and open traverse survey.
5. To have the ability to understand leveling and area of irregular figures.

OUTCOMES:-

After conducting the practical's with different instruments and experiments, students shall be able to:

1. Experimentally verify the polygon law of forces and understand the equilibrium condition of concurrent coplanar forces.
2. Calculate support reactions and unknown forces in a static system using both graphical methods and Lami's theorem.
3. Understand the working principles of chaining instruments and measuring distance, perpendicular and oblique offsets from the given reference line.
4. Understand the principle of prismatic compass, open traverse survey.
5. Compute the RL of different points on the ground by spot leveling and also compute the area and volume of the irregular figure.

List of Demonstration/Experiments:

1. To verify the law of polygon using the universal force table.
2. To determine the support reaction using a graphical method for the simply supported beam.
3. To determine the unknown magnitudes of three concurrent, coplanar, and non-collinear forces acting on a body in equilibrium. (Lami's theorem).
4. To demonstrate the various instruments used in surveying.
5. How to determine the horizontal distance between two points by ranging and chaining and lay the perpendicular and oblique offsets.
6. To demonstrate prismatic compass, measurement of bearings and angles, perform a traverse survey.
7. To demonstrate how to determine RL of different points on the ground, compute RL's by the height of instrument (HI) method and rise and fall method.

RECOMMENDED TEXT BOOKS:

1. Rajput .R.K., *Engineering Mechanics*, Dhanpat Rai and Sons New Delhi.
2. Prasad I. B., *Applied Mechanics*, Khanna Publication New Delhi

3. R. S. Khurmi, *Engineering Mechanics*, , S.Chand Publishing
4. D.S. Bedi, *Engineering Mechanics*, , Khanna Book Publishing Co. (P) Ltd., Delhi
5. R.K. Bansal, *A Textbook of Engineering Mechanics*, , Laxmi Publications
6. Rammamurtham S., *Applied Mechanics*, *Dhanpat Rai and Sons New Delhi*
7. Punmia, B.C., *Surveying*, *Laxmi Publications, New Delhi.*
8. Duggal S.K., *Surveying Vol. 1*, *Tata McGraw- Hill Education New Delhi*

REFERENCE BOOKS:

1. Kumar KL, *Engineering Mechanics*, *Tata McGraw- Hill Education New Delhi*
2. Ferdinand.P. Beer. E, Russell, Jr Johnston., David Mazurek, Philip J Cornwell, *"Vector Mechanics for Engineers: Statics and Dynamics"*, *McGraw – Hill Education New Delhi*
3. Timoshenko, and Young D.H., *"Engineering Mechanics"*, *Tata Mc-Graw Hill Education New Delhi.*
4. Chanchandramouli P.N., *Engineering Mechanics*, *PHI Learning Private Limited New Delhi*
5. Palanichamy *Basic Civil Engineering*, , McGraw Hill
6. Satheesh Gopi *Basic Civil Engineering*, Pearson Publish.

EE 10510: FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
EE10510	Fundamentals of Electrical and Electronics Engineering	3	-	1	3	-	2	30	70	40	60

PRE- REQUISITE: Basic knowledge of science and mathematics at XII level.
COURSE OBJECTIVES:

1. To acquaint students with the basic concepts, elements and properties of electrical circuits and networks.
2. To introduce the concept of phasor and power factor improvement in sinusoidal steady- state analysis.
3. To provide foundational understanding of magnetic circuits, three-phase systems, and the principles, construction and applications of electrical machines.
4. To impart foundational knowledge of semiconductor devices, diodes and bipolar junction transistors and their use.
5. To provide a basic understanding of number systems, Boolean algebra, logic gates, and digital circuits including combinational and sequential logic.

COURSE OUTCOMES:

After completing this course, the student will be able to,

- CO1:** Apply knowledge of science and mathematics to explain electrical circuit component characteristics, formulate various circuit describing mathematical models using basic circuit laws and select appropriate solution technique to find DC excitation responses.
- CO2:** Utilize ac excited steady state circuit analysis concepts in real life problems such as need of power factor correction, power factor improvement methods.
- CO3:** Perform practical analysis of magnetic circuits, single-phase transformers, three-phase power supply, and motor performance to identify suitable motors for specific applications.
- CO4:** Analyze and apply the fundamentals of semiconductor devices, diodes and bipolar junction transistors.
- CO5:** Able to design and analyze basic digital circuits using Boolean functions and flip-flops.



SHRI G. S. INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE
B. TECH. FIRST YEAR (4 YDC)
EE10510: FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS
ENGINEERING

COURSE CONTENTS: THEORY:

Course Contents	CO	PO	BTL
Unit I Voltage and current sources & transformation, properties of resistors, capacitors and inductors, Kirchhoff's laws, Mesh and Nodal analysis, Star-Delta transformations, Superposition theorem, Thevenin's and Norton's theorem, Maximum power transfer theorem.	1	1,2,3 , 4,11	4
Unit II Circuit analysis in sinusoidal steady state, various signals & waveform analysis, Concept of phasor, impedance, admittance, phasor diagrams, Complex, real and reactive power in ac circuits, Power factor and their corrections. Generation of three phase voltages, star and delta connection, Line and Phase quantities, Three-phase power, Three-phase Power measurement techniques.	2	1,2,3 , 4,11	4
Unit III Magnetic circuits: definitions, B-H curves, Magnetic circuit calculations, laws of electro-magnetic induction, Hysteresis and eddy current losses. Single phase Transformer: construction & principle, EMF equation, equivalent circuit, regulation and efficiency. Three phase induction motor: Basic construction, working of three phase induction motor, losses, name plate rating.	3	1,2,3 , 4,11	4
Unit IV Types of semiconductors, PN junction diode and its VI characteristics. Applications of diode as rectifier. Zener diode, its characteristics and its application as voltage regulator. Bipolar junction transistor (BJT), its characteristic in CE mode, Introduction to biasing techniques and use of BJT in power supply.	4	1,2,4, 11	4
Unit V Number system, Binary codes, Boolean algebra, Boolean functions, logic gates. Simplification of Boolean functions using K map (2-3 variable), Combinational logic: Half adder, Full adder. Flip-flops RS, JK, and their applications, Fundamentals of Communication Engineering.	5	1,2,4	4



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B. TECH. FIRST YEAR (4 YDC)
EE10510: FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS ENGINEERING

ASSESSMENT:

- A. Continuous evaluation through two mid-term test with a weightage of 30% of the total marks. It includes class attendance as well as assignments on the course topics.
 B. The end-term theory examination weightage is 70%.

TEXT BOOKS RECOMMENDED:

1. Vincent Del Toro, "Electrical Engineering Fundamentals", Prentice-Hall of India.
2. I. J. Nagarath, "Basic Electrical Engineering", Tata McGraw Hill Publication.
3. Dennis L. Eggleston, Basic Electronics for scientists and engineers, Cambridge Press, 2011
4. Robert Boylestad and Louis Nashelsky, Electronic Devices and Circuit Theory, Prentice Hall Upper Saddle River, New Jersey Columbus, Ohio.

REFERENCES BOOKS:

1. Fitzgerald Higginbotham & Grabel, "Basic Electrical Engineering", Tata McGraw Hill Publication.
2. Joseph Edminister, Mahmood Nahvi "Electric Circuits", Schaum's outlines series.
3. J. B. Gupta, "A Textbook of Basic Electrical and Electronics Engineering", S. K. Kataria & Sons.
4. M. Morris Mano and Michael D. Ciletti, Digital Design, Pearson education, Inc., Upper saddle river, NJ.

CO-PO MAPPING:

CO's	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	1	-	-	-	-	-	-	1	3	1	1
CO2	3	3	3	3	-	-	-	-	-	-	1	3	1	1
CO3	3	3	3	3	-	-	-	-	-	-	1	3	2	1
CO4	3	2	-	2	-	-	-	-	-	-	1	3	3	1
CO5	3	2	-	2	-	-	-	-	-	-	1	3	3	1
Average	3	2.6	3	2.2	-	-	-	-	-	-	1	3	2	1



SHRI G. S. INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE
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EE10510: FUNDAMENTS OF ELECTRICAL AND ELECTRONICS
ENGINEERING

LABORATORY:

LABS/WEEK	PRACTICAL CREDITS	PRACTICAL		TOTAL MARKS
		SW	END SEM	100
2	1	40	60	

OBJECTIVES: The fundamental of Electrical engineering Laboratory is designed:

1. To conduct experiments for the verification of Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL), and to examine the change in resistance with variation in temperature using appropriate measuring instruments and setups.
2. To experimentally determine the power factor of a single-phase load by measuring voltage, current, and real power, and calculating the phase angle between voltage and current to assess power usage efficiency; and to determine the inductance and time constant of a choke coil using voltage-current measurements under varying conditions.
3. To experimentally determine the efficiency of a single-phase transformer under different loading conditions and to calculate the turn's ratio by measuring primary and secondary voltages, comparing with nameplate values for validation.
4. To experimentally test and verify the PN junction diode and Bipolar junction transistor using appropriate laboratory equipment.
5. To experimentally verify the truth table of basic logic gates and able to design new Boolean function.

LABORATORY OUTCOMES:

At the end of this course students will be able to

1. Validate KVL and KCL through experimental analysis and observe the effect of temperature on resistance, enhancing understanding of circuit behavior under varying conditions.



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2. Determine the power factor of a single-phase load and calculate the inductance and time constant of a choke coil using experimental voltage and current measurements.
3. Evaluate transformer efficiency and verify turns ratio through experimental methods using direct loading and voltage measurements.
4. Demonstrate the characteristic of basic electronics components.
5. Verify basic logic gate integrated circuits to solve Boolean functions.

LABORATORY INSTRUCTIONS:

1. In the electrical machine lab, students work with equipment operating on 220V DC, 50Hz single-phase AC, or 400V 50Hz three-phase AC. Since even 110V can be fatal, extreme caution is necessary, as shock risk increases with voltage.
2. Familiarize yourself with the shock-chart instructions, which are provided in the laboratory.
3. Do not touch any terminal or switch without ensuring that it is dead. While performing experiment, make sure to not touch any live terminals.
4. Wearing of shoes with rubber soles is desirable.
5. Make sure that all the electrical connections are right and checked properly by faculty and laboratory assistants, before switching on any circuit. Faulty connections may cause large amount of power, resulting in the damage of parts of the experiment.



SHRI G. S. INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE
B. TECH. FIRST YEAR (4 YDC)
EE10510: FUNDAMENTS OF ELECTRICAL AND ELECTRONICS
ENGINEERING

S. No.	LIST OF EXPERIMENTS	CO	PO	BTL
1	Verification of Kirchhoff's Voltage Law and Kirchhoff's Current Law.	1	1,2,3,4,5,6,7,8,9,11	4
2	To study the variation of resistance with variation in the temperature.	1	1,2,3,4,5,6,7,8,9,11	4
3	To measure the power consumed by a single-phase circuit using a Single-phase wattmeter and computation of load power factor.	2	1,2,3,4,5,6,7,8,9,11	4
4	To determine the inductance and time constant of a choke coil.	2	1,2,3,4,5,6,7,8,9,11	4
5	Determination of efficiency of single-phase transformer by direct loading.	3	1,2,3,4,5,6,7,8,9,11	4
6	To determine the turns ratio of a single-phase transformer.	3	1,2,3,4,5,6,7,8,9,11	4
7	To test the PN junction Diode and experimentally verify its V-I characteristics.	4	1,2,4,11	4
8	To test the BJT and Experimentally verify its input and output characteristics in CE mode.	4	1,2,4,11	4
9	To study logic gates and verify properties of NAND and NOR gates as universal building block.	5	1,2,4	4

10	Simplification and implementation of Boolean functions.	5	1,2,4	4
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ASSESSMENT:

- A. Continuous evaluation of laboratory journals with a weightage of 30%. It includes lab attendance as well as experiments performed in the lab.
- B. The end-term practical examination weightage is 70%.

CO-PO MAPPING:

CO's	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	3	3	3	-	2	3	1	1
CO2	3	3	3	3	2	-	3	3	3	-	2	3	1	1
CO3	3	3	3	3	2	1	3	3	3	-	2	3	2	1
CO4	3	2	-	2	-	-	-	-	-	-	1	3	3	1
CO5	3	2	-	2	-	-	-	-	-	-	1	3	3	1
Average	3	2.6	3	2.6	2	1	3	3	3	-	1.6	3	2	1

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
I Year BE (4YDC)
Course Title: HU10512 : LANGUAGES FOR ENGINEERS

		Credit			Contact Hours			Maximum Marks			
Subject Code	Subject Nomenclature	L	T	P	L	T	P	CW	End Sem	SW	Pr
HU10512	Languages for Engineers	1	1	2	1	1	1	30	70	40	60

PRE-REQUISITE: Nil

COURSE OBJECTIVES:

The primary objective of the Communicative English course is to develop learners' ability to use English effectively and confidently in everyday communication. The course aims to enhance students' speaking, listening, reading, and writing skills, with a strong focus on interaction, fluency, and practical usage.

COURSE OUTCOMES:

After successful completion of the course, students will develop competency in the following areas:

CO1:	To demonstrate effective communication by applying basic grammar, utilizing LSRW skills, employing reading comprehension strategies such as SQ3R, and distinguishing between verbal and nonverbal communication.
CO2:	To develop well-structured and coherent projects by applying the concepts, principles, and formats of project writing, and refining skills through analysis of samples and models..
CO3:	To deliver impactful oral presentations by applying audience analysis, purpose definition, content organization, appropriate medium selection, and effective PowerPoint design techniques.

COURSE CONTENT:

Unit 1 :	Communication Skills:
	(i) Meaning and significance (ii) Basic grammar (iii) The LSRW(Listening, Speaking, Reading, Writing) skills (iv) Reading Comprehension, The SQ3R Technique (v) Types of Communication: Verbal and Nonverbal
Unit II	Project Writing:
	(i) Concept and importance (ii) Features and principles of Project Writing (iii) Structure and Format of Project Writing (iv) Samples and models for practice
UNIT III:	Speaking Skills and Presentation Strategies:

- | |
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| <ul style="list-style-type: none"> (i) Techniques of Oral Presentations, (ii) Defining the Purpose, Analyzing the Audience, Establishing the main idea, Selecting the Appropriate Channel and Medium (iii) Effective Power point presentation |
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BOOKS RECOMMENDED:

1. *High School English Grammar & Composition*, Wren & Martin. Blackie Elt Books, an Imprint of S.Chand Publishing, 2018.
2. Sharma, R.C. and Krishna Mohan, *Business Correspondence and Report Writing*, New Delhi: Tata McGraw-Hill, 2009.
3. Raman, Meenakshi and Sangeeta Sharma, *Technical Communication*, New Delhi: Oxford Univ. Press, 2011.
4. Kapoor A.N., *Business Letters for Different Occasions*, New Delhi: S. Chand & Co. Pvt. Ltd., 2012.

REFERENCES RECOMMENDED:

1. Gupta, Nikhil. *BlackBook of English Vocabulary*. 2024 ed., Nikhil Gupta Publications.
2. Seaton, Anne. *Focus on Grammar: Workbook 1–4*. Marshall Cavendish, 2018.
3. Bhatia, R.C., *Business Communication*, New Delhi: Ane Books Pvt. Ltd., 2012
4. Magan, Sangeeta, *Business Communication*, New Delhi: Biztantra Publications, 2010.

ONLINE RESOURCES:

<https://archive.nptel.ac.in/courses/109/106/109106116/>

<https://www.coursera.org/courses?query=english%20grammar>

OPTIONAL CONTENT

हिंदी

UNIT IV:	साहित्यिक अभिव्यक्तियाँ
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पाठ्यवस्तु:

- माखनलालचतुर्वेदी - *पुष्पकी अभिलाषा* (कविता)
- प्रेमचन्द - *सपनों का दंड* (कहानी)
- भारतेन्दुहरिश्चंद्र - *आचरण की सभ्यता* (निबंध)
- महमूद ताबेबगम - *सौंदर्य की नदीनर्मदा* (यात्रावृत्तांत)
- शरद जोशी - *अफसोस* (व्यंग्य)

गतिविधियाँ:

- कविता का सस्वर वाचन, कहानी वाचन
- निबंध एवं आत्मकथन लेखन का अभ्यास

इकाई V	भाषा संरचना और अभिव्यक्ति
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पाठ्यवस्तु:

- शब्दरचना:उपसर्गएवंप्रत्यय
- शब्दसंकार:तत्सम, तद्भव, देशज, विदेशीशब्द, संकरशब्द, नवीनशब्द
- पारिभाषिकशब्दावली:अनुवादयोग्यपारिभाषिकशब्दोंकेलिएउपयुक्तहिंदीशब्दचयन

संदर्भग्रंथ:

1. प्रेमचंद - मानसरोवर, भाग 1-8
2. डॉ. वासुदेवनंदनप्रसाद - आधुनिकहिंदीव्याकरणऔररचना

OR

SANSKRIT

इकाई IV	संस्कृतभाषाया:स्वरूपम्, साहित्यं च
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पाठ्यवस्तु:

- संस्कृतभाषाया:परिचयः, स्वरूपम्, चमहत्वम्
- वर्णानांउत्पत्तिः, उच्चारणस्थानानिच
- संस्कृतवर्णपरिचयः
- वैदिक-लौकिकसंस्कृतयोःपरिचयः, विशेषताःच
- वैदिकसाहित्यस्यसंक्षिप्तपरिचयः
- शिवसंकल्पसूक्तम् (मन्त्राः 1-6)
- भर्तृहरिणःनीतिशतकात् (पद्याः 1-15)
- अनुवादःचव्याख्या

इकाई V	संस्कृतव्यवहारःतथाभाषाप्रयोगकौशलम्
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पाठ्यवस्तु:

- संस्कृतसंभाषणकौशलम्
- आत्मपरिचयः, विभक्तीनांप्रयोगः, सर्वनाम-शब्दानांचअव्ययस्यप्रयोगः

- सङ्ख्या: - एकतःशतम् (1-100)
- श्लोकपाठः - (पञ्चश्लोकानांपठनम्)
- रचनात्मकःअनुवादः
- *अनुवादकौमुदी*- पाठाः 1-5
- सन्दभग्रन्थः

1.रचनाअनुवादकौमुदी,डॉकफपलद्ववेदी,वशववद्यालयप्रकाशनवाराणसी,२०१९

ASSESSMENT: Through End-Sem. Theory Exam, Theory Sessionals, Mid-SemTests, Assignments.

PY10514: BIOLOGY FOR ENGINEERS

		Credit			Contact Hours			Maximum Marks			
Subject Code	Subject Nomenclature	L	T	P	L	T	P	CW	End Sem	SW	Pr
PY10514	Biology for Engineers	2	0	1	2	0	2	30	70	40	60

Course Title: Biology for Engineers (Theory & Practical)
Duration: One Semester (Approx. 20–24 hours of lectures)
Target Group: 1st & 2nd semester undergraduate engineering students

S. No.	Subject Code	Subject	L	T	P	Theory Credits	Practical Credits	Maximum Marks				
								Th	CW	SW	Pr.	Total
1.		Biology for Engineers	2	-	2	02	01	70	30	40	60	200

Biology for Engineers (Theory)

Course Objectives

1. To introduce foundational biological concepts relevant to engineering.
2. To bridge biology with fields such as electronics, materials, computer science, mechanical and chemical engineering.
3. To develop an appreciation for biomimicry, biosystems, and emerging bio-inspired technologies.
4. To highlight the interface between engineering and Biology for solving real-world problems.
5. To prepare students for interdisciplinary careers in biotechnology, biomedical, environmental, and health sectors.

Course Outcomes (COs)

CO Code	Course Outcome	Bloom's Taxonomy Level
CO1	Understand the structure and function of biomolecules, cells, and enzymes and relate them to biological systems.	Understanding
CO2	Explain the principles of genetics, molecular biology, and genetic engineering and their application in biotechnology.	Understanding / Applying
CO3	Identify the role of microorganisms in health, environment, and industry and apply microbiological techniques in biosafety and bioprocesses.	Applying / Analyzing
CO4	Analyze human physiological systems and their interface with biomedical devices, biomaterials, and sensors.	Analyzing / Applying
CO5	Apply concepts from biomimicry, systems biology, and synthetic biology to develop innovative, bio-inspired engineering solutions.	Applying / Creating

Program Outcomes (POs) (Based on AICTE Graduate Attributes)

PO Code	Program Outcome Description
PO1	Engineering knowledge
PO2	Problem analysis
PO3	Design/development of solutions

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PO Code	Program Outcome Description
PO4	Conduct investigations of complex problems
PO5	Modern tool usage
PO6	The engineer and society
PO7	Environment and sustainability
PO8	Ethics
PO9	Individual and team work
PO10	Communication
PO11	Project management and finance
PO12	Lifelong learning

CO-PO Mapping Matrix

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	-	-	-	-	-	-	-	1
CO2	3	3	2	2	2	1	-	-	-	-	-	2
CO3	3	3	2	2	2	3	3	-	-	-	-	2
CO4	3	3	2	2	3	1	-	1	-	1	-	2
CO5	3	3	3	2	3	2	2	-	1	1	2	3

Bloom's Taxonomy Breakdown per Unit

Unit	Topic Focus	Bloom's Level(s)
Unit I	Cell structure, biomolecules, enzymes	Remembering, Understanding, Applying
Unit II	DNA, gene expression, CRISPR	Understanding, Applying
Unit III	Microbial classification, industry use, biosafety	Applying, Analyzing
Unit IV	Physiology, medical devices, biomaterials	Analyzing, Applying
Unit V	Biomimicry, synthetic biology, systems thinking	Applying, Creating

Unit I – Introduction to Biology (4 hours)

1. Origin and evolution of life, overview of living systems:

Levels of biological organization – Definition of life: metabolism, growth, response to stimuli, reproduction. Hierarchical levels. Importance of **emergent properties**: complex traits arising from simpler components.

2. Cell structure and function: prokaryotes vs. eukaryotes:

- ##### 3. Structure and function of biomolecules: proteins, nucleic acids, lipids, carbohydrates.
- Proteins** - Composed of amino acids (20 types). **Primary to quaternary structure.**
Enzymes, structural proteins, transporters (e.g., haemoglobin).

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14-13C cosy: HETCOR
+ correlates 13C nuclei with directly

4. **Enzymes: mechanisms, kinetics, applications in industry: Industrial Applications - Food**
Industry: amylase in baking, proteases in cheese. **Pharmaceuticals:** enzyme-based drug synthesis. **Textiles and Detergents:** cellulases, lipases. biocatalysts and green chemistry.
Learning Outcomes: Understand biological building blocks and their engineering relevance.

Unit II – Genetics, Molecular Biology, and Applications (4 hours)

1. DNA structure and replication
2. Central Dogma: Transcription and Translation
3. Genetic code and mutations
4. Genetic engineering: recombinant DNA, CRISPR basics
5. Applications in diagnostics, agriculture, and therapeutics

Learning Outcomes: Link molecular biology to tools used in biotechnology and bioinformatics.

Unit III – Microbiology and Industrial Applications (4 hours)

1. Microorganisms: bacteria, viruses, fungi
2. Growth, reproduction, and control of microbes
3. Industrial microbiology: fermentation, biofuels, probiotics
4. Biosafety and sterilization techniques

Learning Outcomes: Recognize how microbes are used in engineering and product development.

Unit IV – Human Physiology and Biomedical Applications (4 hours)

1. Overview of major systems: nervous, cardiovascular, respiratory
2. Tissue engineering and organ systems
3. Sensors and medical devices interfacing with biology
4. Biomaterials and their compatibility

Learning Outcomes: Appreciate the integration of biology in medical and health-tech innovation.

Unit V – Biomimicry, Systems Biology, and Synthetic Biology (4 hours)

1. Biological inspiration for design (e.g., gecko tape, shark-skin coating)
2. Systems biology: modelling biological networks
3. Synthetic biology: gene circuits, biological computing
4. Case studies: bio-robots, bio-inspired drones, DNA computing

Learning Outcomes: Explore how engineering draws from nature for advanced system design.

Suggested Textbooks & References

M. Thirani

1. Arthur T. Johnson, "Biology for Engineers, Second Edition (27 December 2018), CRC Press.
2. Bruce Alberts, Dennis Bray, Julian Lewis, Martin Raff, Keith Roberts, "Molecular Biology of the Cell, Second edition (31 March 1989), Routledge;
3. Biotechnology, B.D. Singh
4. NPTEL Courses (Biology for Engineers by IIT Madras).

Biology for Engineers (Practical)

Course Objectives

1. To develop familiarity with biological laboratory techniques relevant to engineering applications.
2. To reinforce fundamental concepts of biology through practical observation and experimentation.
3. To enhance skills in microscopy, biochemical testing, DNA handling, and microbial techniques.
4. To introduce students to bioinstrumentation and computational tools such as virtual labs or bioinformatics.
5. To encourage application of biological understanding in solving interdisciplinary problems involving biotechnology, biomedical, or environmental technologies.

List of 10 Practicals.

1. Laboratory Safety and Introduction to Biology Lab

- **Objective:** Understand lab safety protocols, biosafety levels, and proper handling of biological specimens.
- **Skills:** PPE usage, waste disposal, good laboratory practices.

2. Use of Microscope and Observation of Cell Types

- **Objective:** Learn how to handle a compound microscope and observe plant and animal cells.
- **Specimens:** Onion peel, human cheek cells.
- **Stains:** Iodine, methylene blue.

3. Study of Cell Organelles Using Permanent Slides or Micrographs

- **Objective:** Identify nucleus, mitochondria, chloroplasts, and other organelles.
- **Activity:** Diagram labeling, organelle function interpretation.

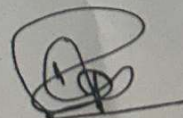
4. Observation of Mitosis in Onion Root Tip

- **Objective:** Observe stages of mitosis (prophase to telophase).
- **Stain Used:** Acetocarmine or Feulgen stain.

M. Tindani

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5. DNA Extraction from Plant Material (e.g., Banana or Spinach)

- **Objective:** Understand basic DNA extraction protocol.
- **Activity:** Precipitation of DNA using alcohol and visualization.

6. Microbial Culture and Gram Staining

- **Objective:** Culture and stain bacteria to differentiate Gram-positive and Gram-negative strains.
- **Application:** Microbial identification and classification.

7. Enzyme Activity Assay (e.g., Amylase on Starch)

- **Objective:** Measure enzyme activity based on color change.
- **Application:** Biocatalysis in food/pharma industries.

8. Biochemical Tests for Biomolecules

- **Tests:**
 - **Benedict's Test** for reducing sugars
 - **Biuret Test** for proteins
 - **Iodine Test** for starch
- **Objective:** Qualitative identification of biomolecules.

9. Study of Human Physiology Using Models and Sensors

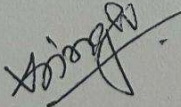
- **Objective:** Demonstrate circulatory and respiratory systems using anatomical models.
- **Optional:** Use of pulse sensor or digital spirometer (if available).

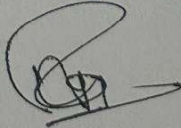
10. Virtual Lab on Bioinformatics or PCR Simulation

- **Objective:** Introduction to tools like NCBI BLAST, genome browsers, or virtual PCR.
- **Resources:** NPTEL, Amrita Virtual Labs, DNA Learning Center.

M- Tiwari
(Prof. M. Tiwari)

(Ms. Avni Jain)


(Dr. Vibeet Singh)


(Prof. R. Khatri)

IP10584: DESIGN THINKING AND MANUFACTURING PRACTICES

Subject Code	Subject Nomenclature	Credit			Contact Hours			Maximum Marks			
		L	T	P	L	T	P	CW	End Sem	SW	Pr
IP10584	Design Thinking and Manufacturing Practices	0	1	2	0	1	4	-	-	40	60

COURSE OBJECTIVES:

1. To think outside of the box and generate multiple ideas and develop innovative solutions to complex problems.
2. To create conceptual design, solutions and testing them with users to gather feedback and iterate on their designs.
3. To be conversant with different carpentry and foundry tools, operations, and procedures.
4. To be conversant in different welding and forging methods.
5. To be familiar with different advanced and conventional machining processes.

COURSE CONTENTS : THEORY AND PRACTICE:

UNIT:1

Introduction to Design thinking and Innovation, Models of design thinking and Innovation, process and tools, mind mapping and project selection, brainstorming, primary and secondary research, information data analysis.

UNIT 2:

Ideation, Project brainstorming, Lateral thinking, Synectics, Analogical Thinking, Metaphors, Nature Inspiration, Concept Evaluation, Concept-Maps, soft medium and final prototyping, Testing.

UNIT 3 :

(A) Wood working: Applications of wood/timber as an engineering material, natural wood, their types, hard and soft wood. Introduction to wood working tools for various operations like saw, chisel, jack plane, marking and measuring tools. Types of joints and their use. Manufacture of plywood and ply-boards etc. Preservation of timber, glues and adhesives in wood working, use of nails, screw etc., introduction to paints and varnishes for protection and decorative purposes.

(B) Foundry Process: Introduction to moulding and casting processes. Moulding sand and its main constituents, moulding tools and their use, green sand moulding practice. bench and pit moulding, properties of moulding sand, use of additives in moulding sand, use of patterns and their types, pattern materials. Use of cores and chaplets. Pouring of metal and removal of castings. Casting defects and their remedies.

UNIT:4

(A) Forging Process: Hot and cold working of metals, material suitable for forging operation, simple forging operation with use of tools like hammers, anvil, swage block, dies, tongs, punches, chisel, drifts etc. Forging operations like upsetting, bending and drawing out etc.

(B) Welding Process: Introduction to welding practice and classification (fusion and pressure welding). Types of electrodes and shielding of arc. Manual Metal Arc Welding (MMAW). AC and DC welding, tools and equipment used in electric arc welding. Types of welded joints, joints preparation and welding positions. Gas welding - Oxyacetylene welding. High- and low-pressure gas welding. Types of flames and their use. Gas cutting. Defects in welding.

UNIT:5

(A) Machining process: Introduction to chip removal processes, working of simple machine tools like lathe, drill, shaper, planer. Single point cutting tool and their materials and various angles. Simple lathe operations like turning, facing, taper turning, boring, threading, knurling etc. Use of taps and dies, drills and center drills and reamers on lathe. Quick return mechanism used in shaper and planner and simple operations on these machine tools. Idea Lab: CO2 laser- 2D Cutting and engraving, CNC Router- 3D engraving and cutting, Pratham 2.0 3D Printer- Rapid Prototyping, 3D Scanner, PCB prototype.

(B) Fitting Practice: Need of fitting shop, common measuring and marking tools like Engineer's scale, try square, surface plate, marking gauge, V-block, center punch. Use of inside, outside and odd leg callipers, micrometers and verniers, combination set, bevel protector. Use of files, their types in respect of length and cross section, single and double cut file, filing techniques. Use of hacksaw, drills, tapes and dies.

TEXT BOOK RECOMMENDED:

1. Eli Woolery “Design Thinking Handbook” DesignBetter.co by Invision
2. R.S. Khurmi and J.K. Gupta “Workshop Technology”, S. chand.
3. S.K Hajra Choudhary “Elements of Workshop Technology”, Media Promoter & Publishers Pvt. Ltd.

CO-PO MAPPING:

CO's	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO11	PO12
CO1	2	2	2	1	1	1	1	2	1	1	2	1

CO2	2	2	2	2	1	1	1	2	1	1	2	1
CO3	2	1	1	-	1	1	1	1	1	-	1	1
CO4	2	1	2	-	1	1	1	1	1	-	1	1
CO5	2	1	2	-	1	1	1	1	1	-	1	1

SHRI G. S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE

DEPARTMENT OF INDUSTRIAL & PRODUCTION ENGINEERING

B.Tech. First Year (4ydc)

IP10584: Design Thinking and Manufacturing Practices

Section	Branch	Faculty Senior	Faculty Junior
F	EI	Mr. M. Mahajan	Dr Kailash Rai
G	IP	Dr. G.D Thakar	Dr. Kumar Rohit
H	IT	Dr. S. Balsara	Mrs. Angurbala Jayant
I	ME	Mr. S. Mujalda	Dr Kailash Rai
J	BMIL	Dr. A Modi	Mrs. Angurbala Jayant